

— SAME INDUSTRY. DIFFERENT POSSIBILITIES.

From operational data to real climate impact.

NeoCarbon is a specialized GHG accounting platform built specifically for the oil & gas industry. It turns raw field activity data into engineering-grade greenhouse gas insights, regulatory-ready audit reports, and a transparent roadmap to lower your carbon footprint.

More Than Data. Real Control.

Empower operators and executives with defensible numbers.

✓ ONE
PLATFORM✓ FULL
VISIBILITY✓ REAL
ACTION

NEOCARBON ENTERPRISE OVERVIEW

ASSETS: 14 • REAL-TIME FEED

TOTAL EMISSIONS

2,458,732 tCO₂e
▲ 12% vs LY (Audited)

METHANE INTENSITY

0.42 tCO₂e/boe
Target < 0.20%

CARBON INTENSITY

16.8 tCO₂/boe
Top Decile E&P

EMISSIONS BY SCOPE TOTAL BREAKDOWN

Scope 1:	Scope 2:	Scope 3:
62%	12%	26%

Field Alpha (CPF-1)	28%
Field Bravo (LNG Hub)	21%
Field Charlie (Offshore)	17%

BUILT FOR THE REALITIES OF OIL & GAS

PURPOSE-ENGINEERED • NOT GENERIC ESG SOFTWARE



01

Accuracy You Can Trust

Engineering formulas, custom fuel compositions, and thermodynamic engines — not simplistic default factors.



02

OGMP 2.0 Ready

Dedicated methane emissions tracking, source-level reporting, and continuous improvement workflow.



03

All Your Data. One Place.

From wells to terminals, covering every asset, processing facility, and emission source across the chain.



04

More Insights. Less Uncertainty.

Scenario analysis, trends, intensity metrics, and forward-looking decarbonization planning tools.



05

Designed for Low-Carbon Future

Meet regulatory demands, improve operational efficiency, and stay ahead in the global energy transition.

FULL OIL & GAS COVERAGE: FROM WELLHEAD TO FLARE

END-TO-END VALUE CHAIN SCOPE 1, 2 & 3 MAPPING



Upstream Production & Extraction

- › Wells & artificial lift systems
- › Gathering lines & manifolds
- › Gas compression & separation
- › Central Processing Facilities (CPF)



Midstream Transport & Storage

- › High-pressure gas pipelines
- › Booster compressor stations
- › Crude & condensate tank farms
- › LNG terminals & marine loading



Downstream Refining & Distribution

- › Refineries & catalytic crackers
- › Petrochemical processing plants
- › Product storage & blending
- › Truck, rail & shipping logistics



All Emission Sources Speciated & Reconciled

- › Stationary fuel combustion
- › Routine & emergency flaring
- › Venting (tanks, dehy, pneumatics)
- › Fugitives (valves, pumps, LDAR)

THE NEOCARBON ADVANTAGE: 6 FUNCTIONAL MODULES

ENGINEERED FOR ENERGY PROFESSIONALS



Advanced Calculation Engine

TIER 1-2-3

API Compendium 2021 algorithms, real gas compositions, stoichiometric equations, and regional basin factors.



Comprehensive Reporting

ISO 14064

Automated Scopes 1, 2 & 3 reporting aligned with ISO 14064, GHG Protocol, API standards, and corporate ESG formats.



Scenario & Abatement Tools

MACC ANALYSIS

What-if modeling, flare gas monetization payback, VRU capital returns, and low-carbon transition pathways.



OGMP 2.0 & Methane Intel

GOLD LEVEL 5

Reconcile source-level bottom-up physics with top-down measurements (drones, aircraft, satellites) and LDAR campaigns.



Data Integration & Ingestion

AIR-GAPPED

Direct SCADA/historian connectors, secure API pipelines, bulk Excel/CSV ingestion, and verified operator entry workflows.



Full Lineage & Traceability

AUDIT TRAIL

Immutable input-to-result calculation lineage, version-controlled factor provenance, and 100% audit reproducibility.

Your Emissions Data Deserves A Smarter Platform.

Stop relying on fragile spreadsheets. Move to engineering-grade greenhouse gas intelligence.

EXPLORE WHAT'S POSSIBLE → PAGE 01 / 02

Not Just A Reporting Tool. An Emissions Calculation Engine.

NeoCarbon connects operational activity telemetry to engineering-grade greenhouse-gas calculations, speciated methane accounting, and verifier-ready decision support across the entire upstream, midstream, and downstream value chain.

☆ ONE CALCULATION ARCHITECTURE

MULTI-TIER ENGINE

Tier 1: DefaultsTier 2: Basin FactorsTier 3: Engineering

Dynamically computes emissions using IPCC default factors, regional basin benchmarks, or thermodynamic process formulas depending on operational data fidelity.

🕒 SOURCE-LEVEL DETAIL

FULL LINEAGE

Well / ComponentBattery / FacilityCorporate Rollup

Maintains the complete, unbreakable chain of custody between raw input parameters, calculation methodology, factor provenance, and reported emissions.

WHAT MAKES NEOCARBON DIFFERENT

THE 6 ENGINEERING PILLARS

01 / CALCULATION
Engineering Formulas

Fluid composition, operating pressures, and physical properties drive process math instead of generic factors.

02 / FACTORS
Flexible Hierarchy

Corporate custom factors cascade into API, regional, and IPCC tables with unit conversion logic.

03 / TRACEABILITY
Input → Math → Result

Every metric preserves full calculation lineage so reported tonnes trace back to raw meters.

04 / METHANE
OGMP 2.0 Workflow

Quantify methane at component level for OGMP Level 4/5 reporting, tracking, and leak mitigation.

05 / COMPOSITION
Custom Gas Speciation

Enter lab chromatography (C1-C6+, CO₂, N₂, H₂S) to compute constituent-specific mass emissions.

06 / CONTROL
Audit Version History

Tracks every modification to activity data, factors, and formulas for verifier review and reproducibility.

FROM RAW DATA TO DECISION-READY INTELLIGENCE: THE CALCULATION PIPELINE

API COMPENDIUM 2021 • IPCC AR5/AR6 GWP HORIZONS

1. Input Data

Meters, SCADA, fuel volumes, run hours, gas analyses

2. Unit Conversion

Sm³, Mcf, bbl, GJ, tonnes, operating P/T normalization

3. Tier Dispatcher

Selects Tier 1 default, Tier 2 basin, or Tier 3 physics

4. Engineering Math

Applies stoichiometry, flash EOS, DRE & custom factors

5. Speciated GHG

Distinct mass totals for CO₂, CH₄, and N₂O

6. Total CO₂e

Carbon & methane intensities, Scopes 1-2-3 & reports

BUILT AROUND OIL & GAS EMISSION SOURCES

EXHAUSTIVE PROCESS & EQUIPMENT COVERAGE

🔥 Stationary Combustion

BoilersGas TurbinesFurnacesGeneratorsProcess Heaters

Computes fuel volume, custom heating value, and combustion efficiency.

⚡ Flaring & Venting

Routine FlareNon-RoutineAssociated GasLiquids Unloading

Models Destruction and Removal Efficiency (DRE), gas composition, and flare tip physics.

🔍 Fugitive Emissions

ValvesCompressorsPump SealsFlangesLDAR

Component-level inventories and methane screening campaign workflows.

🕒 Complete Chain

ProductionProcessingTransportRefiningScope 2 & 3

Organize assets, joint ventures, and facilities across global operations.

Capability / Feature	NeoCarbon Platform	Other Industry Software	Specialized Methane Tools	Spreadsheets
Full Oil & Gas Coverage (Upstream → Downstream)	✓ Comprehensive	~ Partial	~ Partial	X Fragmented
Engineering Formulas & Custom Factors (Tier 3)	✓ Full Physics Engine	~ Partial	X Not Supported	X Broken Formulas
OGMP 2.0 Ready (Tier 1/2/3 & Gold Standard)	✓ Level 1-5 Built-in	~ Partial	~ Partial	X Infeasible
Methane Source-Level Tracking & LDAR	✓ Equipment Granular	~ Partial	~ Limited	X Not Supported
Regulatory Reporting (ISO 14064, API, EU, CBAM)	✓ 1-Click Dossiers	~ Partial	X Not Supported	X Audit Failure Risk
Scenario & Mitigation Planning (MACC Curves)	✓ What-If Simulation	~ Limited	X Not Supported	X Manual / Static
Real-Time / Continuous Monitoring & SCADA	✓ Native Ingestion	~ Limited	~ Partial	X Manual Entry Only
Integrated Dashboard & Intensity Analytics	✓ CO ₂ e/boe & Methane	~ Partial	~ Partial	X Disconnected
Data Sovereignty & Air-Gapped Deployment	✓ 100% On-Premise Air-Gap	~ Limited (SaaS only)	~ Limited	~ Unsecured Files

OGMP 2.0: MORE THAN A STANDARD

Be ready for what's next. NeoCarbon accelerates your journey to Gold Standard compliance:

1. Track Methane at Source

2. Meet UNEP Requirements

3. Improve Transparency

4. Reduction Roadmap

VERIFIER-READY COMPLIANCE & STANDARDS

100% AUDIT REPRODUCIBILITY

ISO 14064-1 • 14064-2OGMP 2.0 Gold StandardAPI Compendium 2021EU Methane Reg 2024/1787EU CBAM Embedded GHGGHG Protocol Corporate

Generates fully auditable calculation dossiers with cryptographic verification seals, reducing third-party auditor hours by up to 80% while shielding against cross-border regulatory penalties.

Deploy Sovereign Carbon Accounting in 15 Minutes

Provide a monthly facility report or fuel analysis. Our team will model your site live and demonstrate your exact intensity KPIs on the spot.

Schedule Demo: contact@neocarbon.dzwww.neocarbon.dz